

MODULE 17

SQL Common Table Expressions CTE

DBMS with SQL - Enterprise Learning Series

ARSHITH LMS | Professional Edition

SQL Common Table Expressions CTE

What is a CTE?

Temporary named result using WITH.
Exists only during query execution.
Benefits: readable, recursive, reusable within query.

```
WITH name AS (SELECT ...) SELECT * FROM name;
```

Simple CTE

```
WITH avg_gpa AS (  
  SELECT AVG(gpa) AS average FROM students  
)  
SELECT s.first_name, s.gpa  
FROM students s, avg_gpa  
WHERE s.gpa > avg_gpa.average;
```

Multiple CTEs

```
WITH  
  dept_avg AS (  
    SELECT dept, AVG(gpa) avg FROM students GROUP BY dept  
  ),  
  top AS (  
    SELECT * FROM students WHERE gpa > 3.8  
  )  
SELECT t.name, d.avg  
FROM top t JOIN dept_avg d ON t.dept = d.dept;
```

Recursive CTE

```
WITH RECURSIVE org AS (  
  SELECT emp_id, name, manager_id, 0 AS level  
  FROM employees WHERE manager_id IS NULL  
  UNION ALL  
  SELECT e.emp_id, e.name, e.manager_id, o.level+1  
  FROM employees e JOIN org o ON e.manager_id = o.emp_id  
)  
SELECT * FROM org ORDER BY level;
```

CTE vs View vs Subquery

CTE: query-scoped, readable, recursive capable
Subquery: inline, not reusable
View: persistent, reusable across sessions

Use CTE for complex single-query logic.
Use View for multi-session reusable logic.